

# Urban Green Infrastructure and Ecosystem Service Deficit Mapping — Tartu, Estonia

## *Spatial Assessment of Green-Infrastructure Capacity, Demand, Deficit and Planning-Priority Zones*

### Urban Green Infrastructure Planning Priority Zones

FIGURE 5

Spatial prioritization for green infrastructure intervention and ecosystem service improvement — Tartu, Estonia

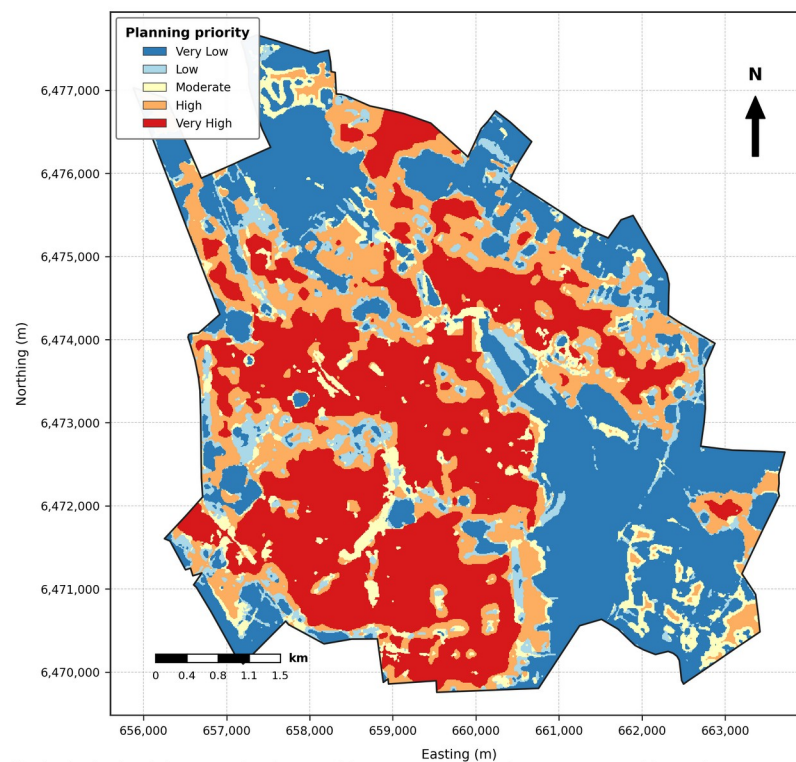


Figure 5. Planning-priority classification showing the relative urgency for urban green infrastructure protection, enhancement or targeted intervention.

Projection: EPSG:3301 — Estonian Coordinate System of 1997  
Spatial resolution: 10 m | Study area: 38.888 km<sup>2</sup>

Data: Project-derived ecosystem service deficit and planning priority model  
Prepared by: Abdullah Abdazeez Ayomide | Project year: 2025

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## Executive Summary

The validated compact urban study area covers 38.888 km<sup>2</sup>. Built-up land is the dominant mapped surface, occupying 21.426 km<sup>2</sup>, equivalent to 55.10% of the study area. Vegetated urban green infrastructure—including dense urban forest, urban woodland, healthy green space and stressed green space—covers 16.619 km<sup>2</sup> or 42.74% of the area.

The ecosystem-service planning-priority assessment identifies 20.230 km<sup>2</sup>, or 52.02% of the study area, as High or Very High priority. Very High priority zones alone occupy 12.092 km<sup>2</sup>, representing 31.09% of the mapped area.

The spatial concentration of urgent planning need is strongly associated with built-up surfaces. Approximately 19.955 km<sup>2</sup> of built-up land falls within High or Very High priority zones. This represents 93.14% of all mapped built-up land and 98.64% of the total High and Very High priority area.

Mean ecosystem-service deficit increases systematically from 0.104 in Very Low priority areas to 0.858 in Very High priority areas. The Very High priority mean is approximately 8.3 times the mean recorded in Very Low priority areas, demonstrating a clear and internally consistent planning gradient.

## Project at a Glance

| Indicator                             | Result                                           |
|---------------------------------------|--------------------------------------------------|
| Study area                            | 38.888 km <sup>2</sup>                           |
| Built-up area                         | 21.426 km <sup>2</sup> (55.10%)                  |
| Vegetated UGI                         | 16.619 km <sup>2</sup> (42.74%)                  |
| High and Very High priority           | 20.230 km <sup>2</sup> (52.02%)                  |
| Built-up land in High/Very High zones | 19.955 km <sup>2</sup> (93.14% of built-up land) |
| Mean ecosystem-service deficit        | 0.514                                            |

# 1. Introduction

Urban green infrastructure comprises interconnected natural, semi-natural and designed green and blue spaces that support ecological functions and contribute to urban environmental quality. In compact urban areas, the distribution of green infrastructure is often uneven, producing spatial differences in ecosystem-service capacity and access to environmental benefits.

This project evaluates the spatial relationship between urban green infrastructure, ecosystem-service capacity, ecosystem-service demand and the resulting service deficit within the compact urban area of Tartu, Estonia. The analysis provides a reproducible planning framework for identifying locations where green-infrastructure protection, enhancement or new intervention should receive priority.

The final planning-priority surface is intended as a strategic decision-support layer. It indicates relative intervention urgency across the study area but does not replace detailed neighbourhood assessment, land-ownership investigation, engineering analysis or field verification.

## 2. Project Objectives

- Classify and quantify major urban green-infrastructure and surface-cover classes.
- Model the spatial distribution of ecosystem-service capacity.
- Represent spatial variation in ecosystem-service demand.
- Identify ecosystem-service deficit through capacity–demand comparison.
- Classify intervention need into five planning-priority categories.
- Generate validated maps, statistics, planning actions and portfolio-ready outputs.

## 3. Study Area

The analysis covers the validated compact urban area of Tartu, Estonia. The authoritative vector boundary has an area of 38.888 km<sup>2</sup>. Rasterization at the project resolution produced an area of 38.887 km<sup>2</sup>, differing from the vector area by less than 0.01%. The common valid comparison area used for analyses requiring complete capacity, demand, deficit and priority data is 38.444 km<sup>2</sup>.

## 4. Data and Methodology

### 4.1 Spatial framework

All analytical raster layers were harmonized to a common 10-metre grid and validated for identical coordinate reference system, dimensions, affine transform, bounds and spatial resolution. The authoritative compact urban boundary was rasterized to create the final production mask.

### 4.2 Urban green-infrastructure classification

The final classification distinguishes Water, Dense Urban Forest, Urban Woodland, Healthy Green Space, Stressed Green Space, Built-up and Bare Surface. Class areas were calculated from validated pixel counts within the authoritative study boundary.

### 4.3 Ecosystem-service capacity

A continuous ecosystem-service capacity score was developed on a 0–100 scale. Higher values indicate greater modelled capacity to provide the ecosystem services represented in the project framework.

### 4.4 Ecosystem-service demand

A normalized demand index was used to represent spatial variation in the need for ecosystem services. The final demand surface ranges from 0 to 1.

#### **4.5 Ecosystem-service deficit**

The deficit index represents the spatial mismatch between capacity and demand. Higher values identify locations where demand is strong relative to existing ecosystem-service capacity.

#### **4.6 Planning-priority classification**

The deficit assessment was translated into five intervention classes: Very Low, Low, Moderate, High and Very High. The classes provide a relative planning hierarchy rather than fixed regulatory zones.

#### **4.7 Validation and reproducibility**

Each production stage included file-existence checks, percentage-total validation, spatial alignment checks, raster-mask validation, output metadata and cross-tabulation controls. Final maps were exported as 300-DPI PNG files and corresponding PDF files.

# 5. Results

## 5.1 Urban Green Infrastructure Distribution

### Urban Green Infrastructure Classification

FIGURE 1

Distribution of water, vegetation, built-up land and other urban surface classes — Tartu, Estonia

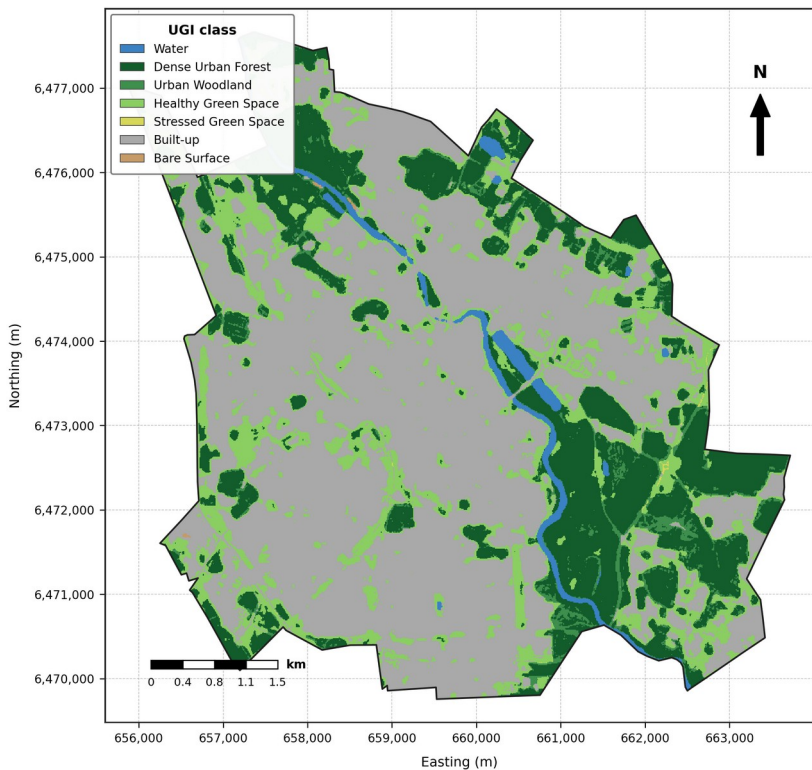


Figure 1. Spatial distribution of the seven urban green infrastructure and land-surface classes within the compact Tartu study area.

Projection: EPSG:3301 — Estonian Coordinate System of 1997  
Spatial resolution: 10 m | Study area: 38.888 km<sup>2</sup>

Data: Sentinel-2 SR Harmonized, Dynamic World V1 and project-derived classification  
Prepared by: Abdullah Abdazeez Ayomide | Project year: 2025

Figure 1. Urban green infrastructure classification within the Tartu compact urban study area.

| UGI class            | Area (ha) | Area (km <sup>2</sup> ) | Share (%) |
|----------------------|-----------|-------------------------|-----------|
| Water                | 82.79     | 0.828                   | 2.13      |
| Dense Urban Forest   | 854.92    | 8.549                   | 21.98     |
| Urban Woodland       | 129.51    | 1.295                   | 3.33      |
| Healthy Green Space  | 675.89    | 6.759                   | 17.38     |
| Stressed Green Space | 1.62      | 0.016                   | 0.04      |
| Built-up             | 2142.60   | 21.426                  | 55.10     |
| Bare Surface         | 1.39      | 0.014                   | 0.04      |

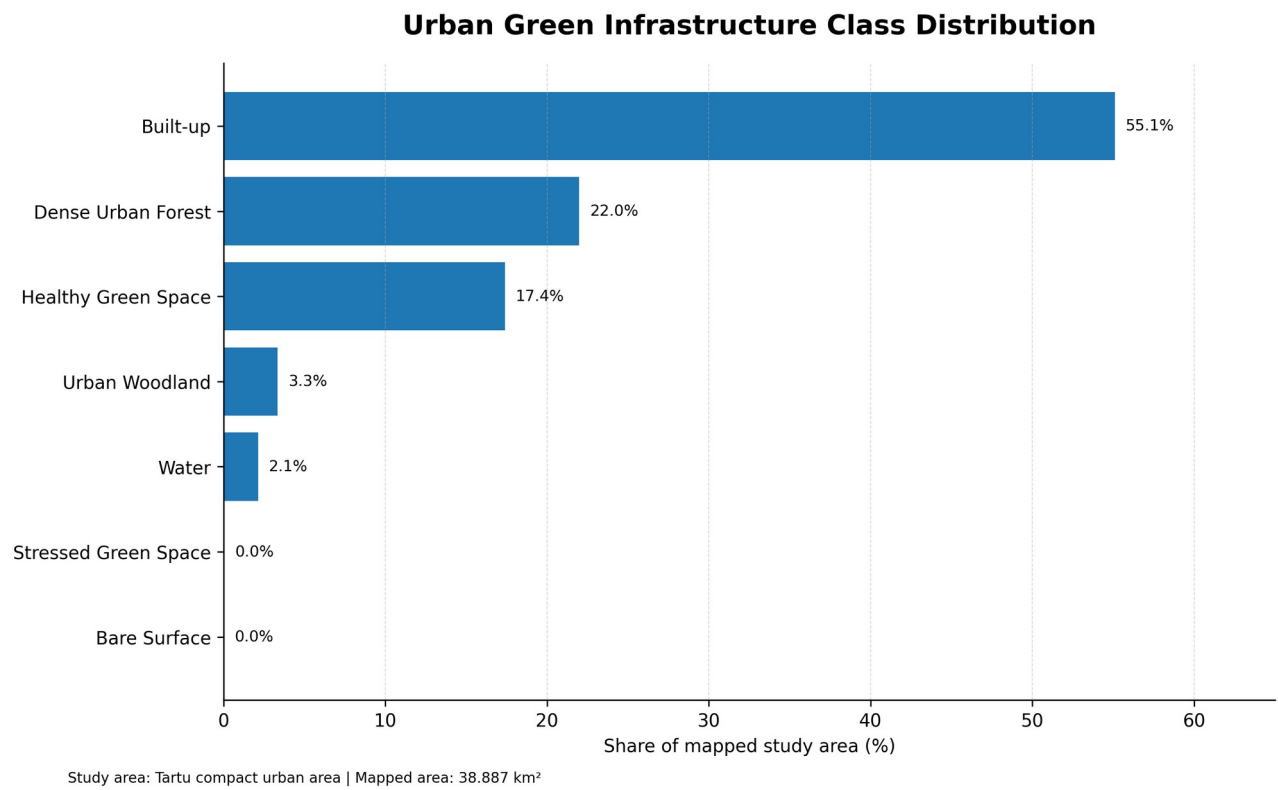


Figure 2. Percentage distribution of urban green-infrastructure classes.

## 5.2 Ecosystem Service Capacity

### Ecosystem Service Capacity

FIGURE 2

Spatial capacity of urban green infrastructure to provide ecosystem services — Tartu, Estonia



Figure 2. Ecosystem service capacity scores, where higher values identify areas with stronger potential to provide urban ecological functions and benefits.

Projection: EPSG:3301 — Estonian Coordinate System of 1997  
Spatial resolution: 10 m | Study area: 38.888 km<sup>2</sup>

Data: Project-derived capacity model based on urban green infrastructure characteristics  
Prepared by: Abdullah Abdazeez Ayomide | Project year: 2025

Figure 3. Modelled ecosystem-service capacity across the study area.

# 5.3 Ecosystem Service Demand

## Ecosystem Service Demand

FIGURE 3

Relative spatial demand for ecosystem services across the compact urban area — Tartu, Estonia

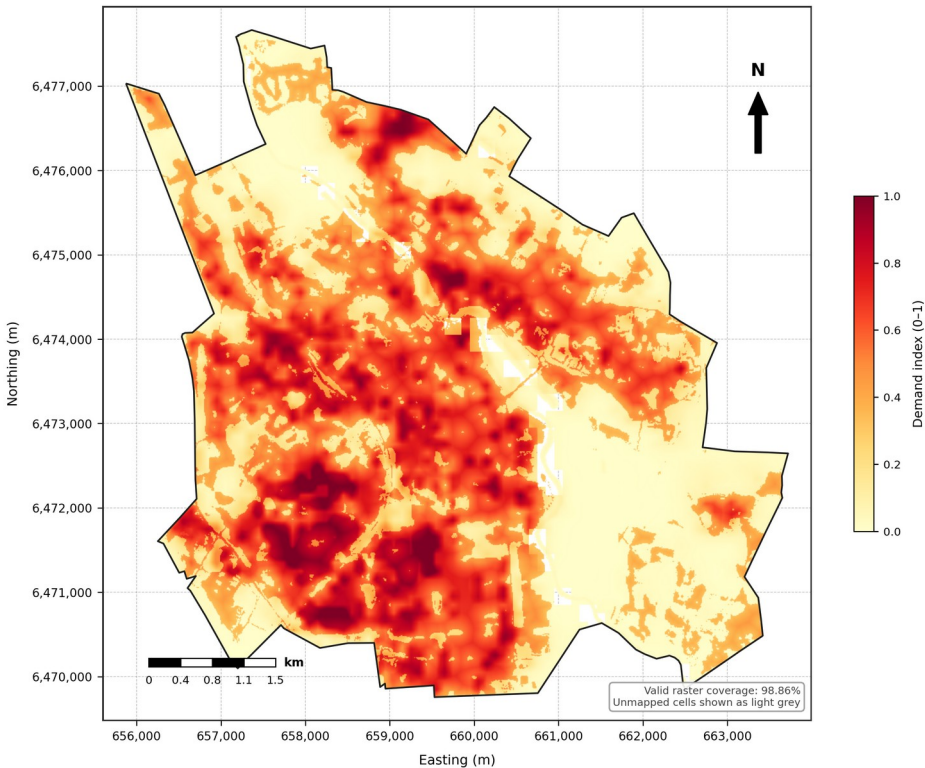


Figure 3. Normalized ecosystem service demand index, with higher values representing areas of greater population, environmental or urban-service need.

Projection: EPSG:3301 — Estonian Coordinate System of 1997  
Spatial resolution: 10 m | Study area: 38.888 km<sup>2</sup>

Data: WorldPop and project-derived urban demand indicators  
Prepared by: Abdullah Abdazeez Ayomide | Project year: 2025

Figure 4. Normalized ecosystem-service demand across the study area.

# 5.4 Ecosystem Service Deficit

## Ecosystem Service Deficit

Mismatch between ecosystem service capacity and demand — Tartu, Estonia

FIGURE 4

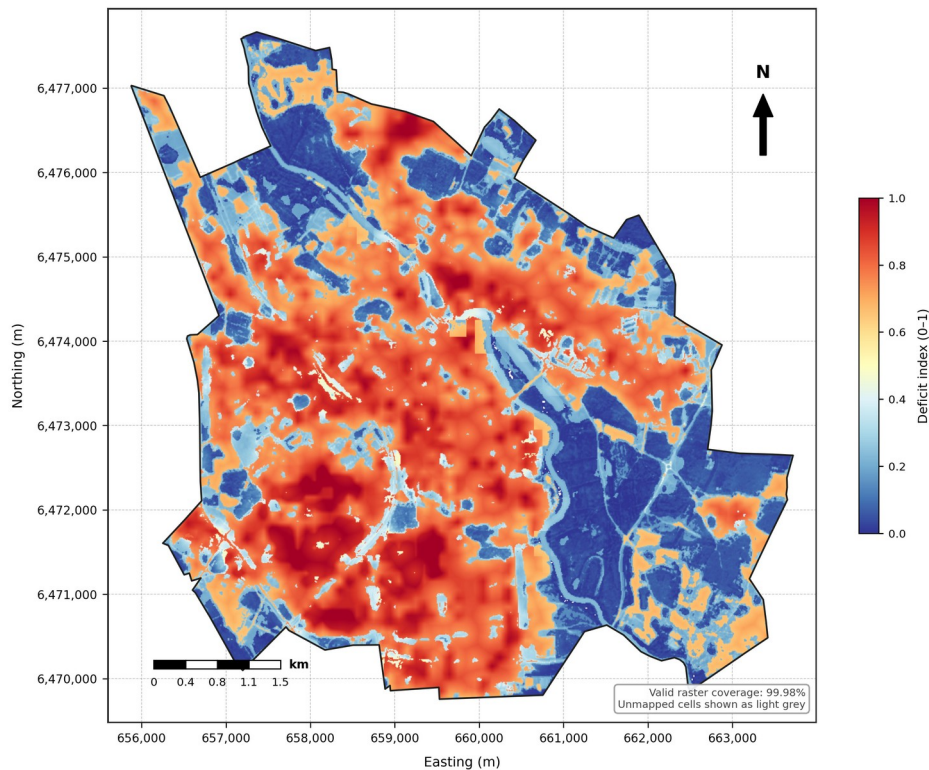


Figure 4. Normalized ecosystem service deficit index, where higher values indicate locations in which service demand is poorly matched by existing ecosystem service capacity.

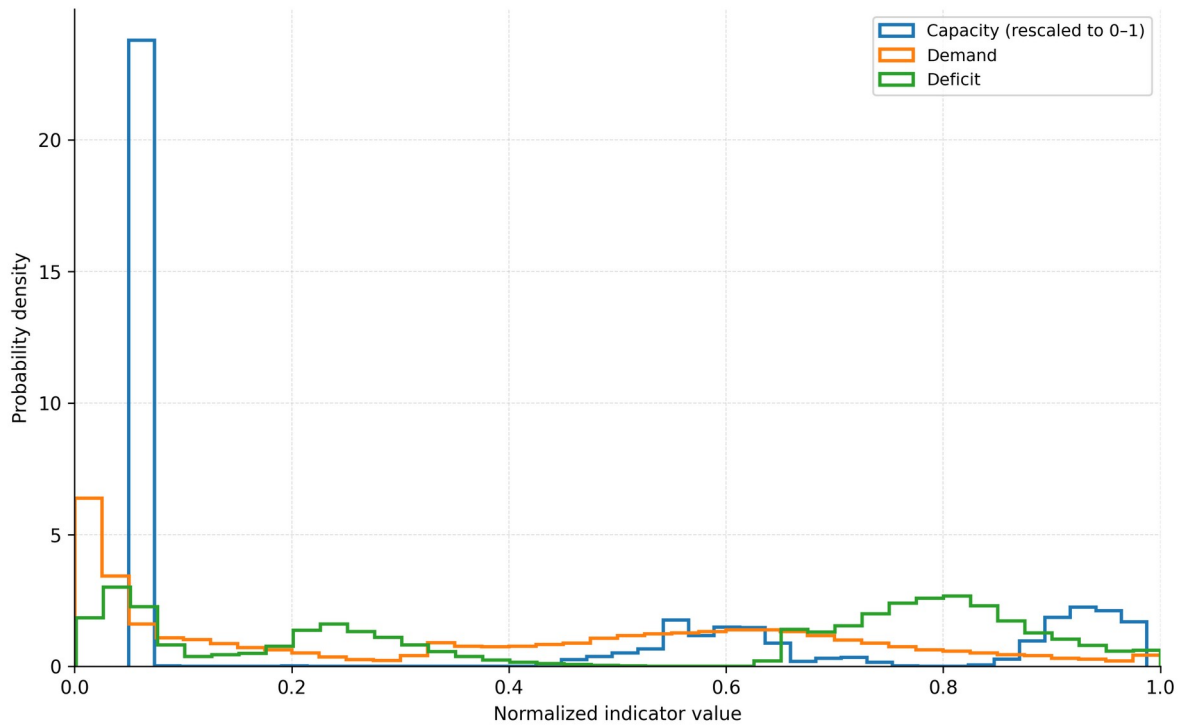
Projection: EPSG:3301 — Estonian Coordinate System of 1997  
Spatial resolution: 10 m | Study area: 38.888 km<sup>2</sup>

Data: Project-derived integration of ecosystem service capacity and demand surfaces  
Prepared by: Abdullah Abdazeez Ayomide | Project year: 2025

Figure 5. Spatial mismatch between ecosystem-service capacity and demand.

| Indicator                  | Minimum | Median | Mean   | Maximum | Standard deviation |
|----------------------------|---------|--------|--------|---------|--------------------|
| Ecosystem Service Capacity | 5.000   | 5.000  | 36.710 | 98.750  | 37.126             |
| Ecosystem Service Demand   | 0.000   | 0.406  | 0.379  | 1.000   | 0.298              |
| Ecosystem Service Deficit  | 0.000   | 0.685  | 0.514  | 1.000   | 0.334              |

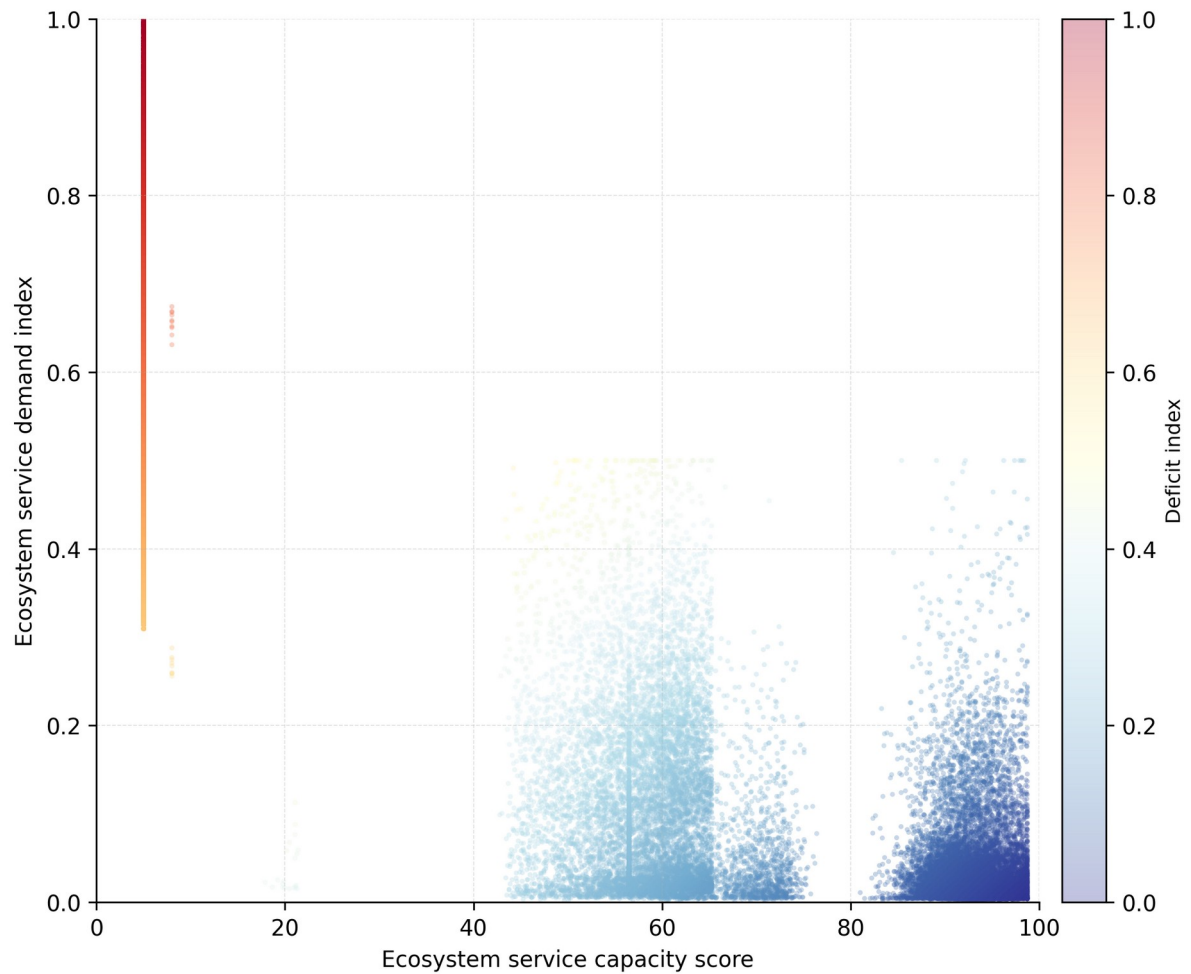
## Distribution of Ecosystem Service Indicators



Common comparison area: 38.444 km<sup>2</sup>

Figure 6. Statistical distributions of ecosystem-service capacity, demand and deficit.

## Relationship Between Ecosystem Service Capacity and Demand



Displayed random sample: 50,000 cells | Common valid population: 384,441 cells

*Figure 7. Relationship among ecosystem-service capacity, demand and deficit.*

# 5.5 Planning-Priority Zones

## Urban Green Infrastructure Planning Priority Zones

FIGURE 5

Spatial prioritization for green infrastructure intervention and ecosystem service improvement — Tartu, Estonia

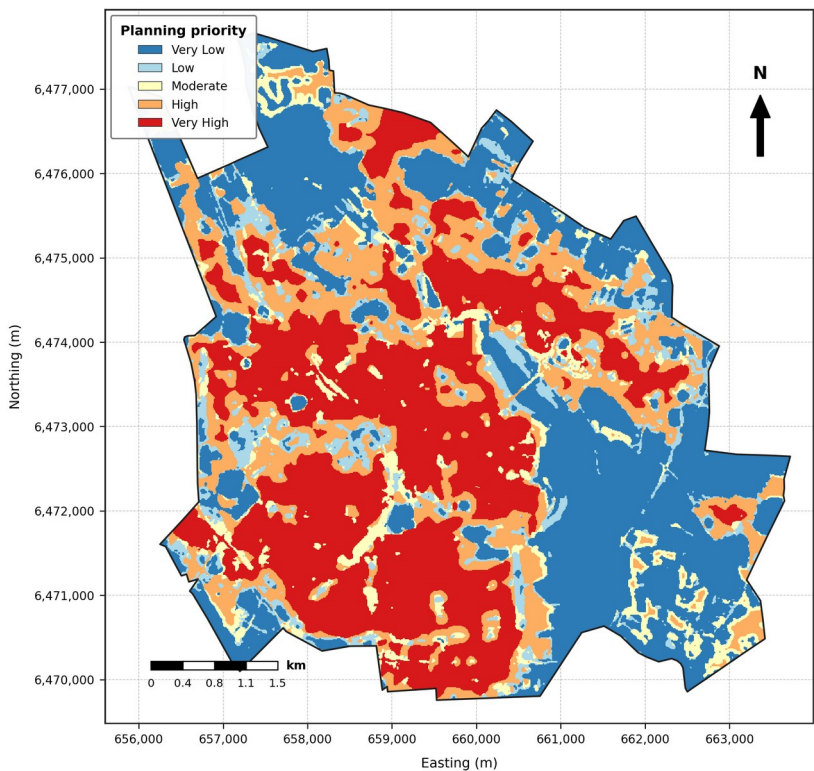


Figure 5. Planning-priority classification showing the relative urgency for urban green infrastructure protection, enhancement or targeted intervention.

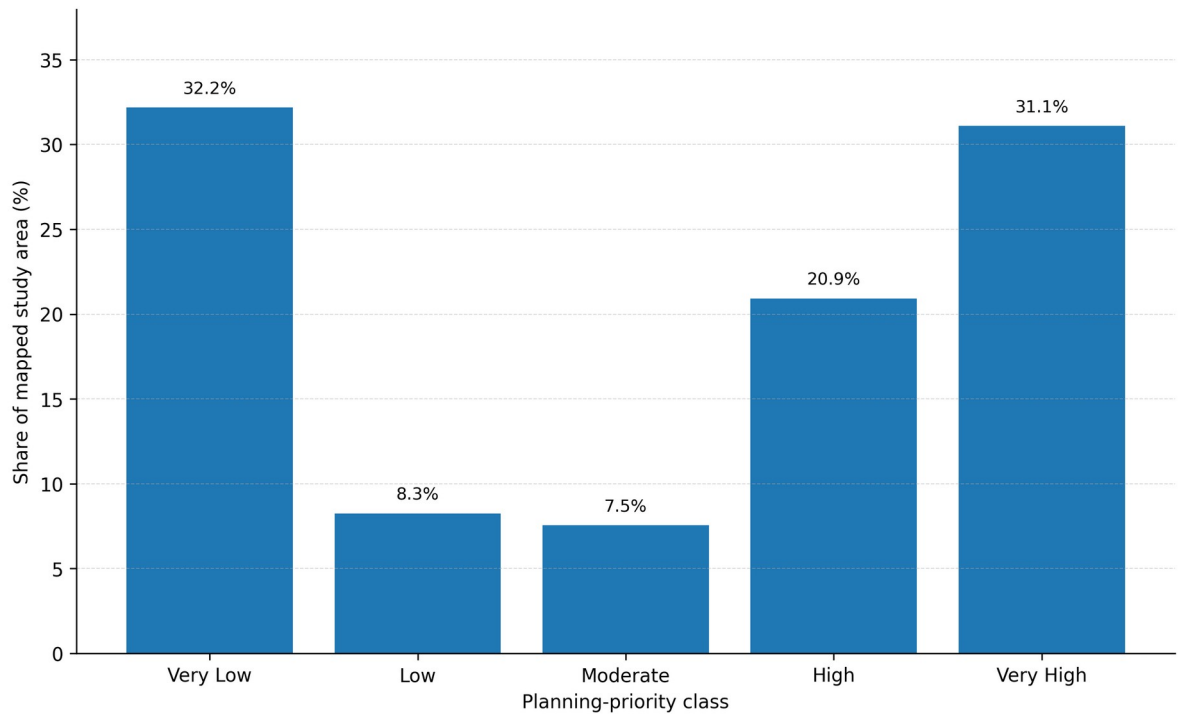
Projection: EPSG:3301 — Estonian Coordinate System of 1997  
Spatial resolution: 10 m | Study area: 38.888 km<sup>2</sup>

Data: Project-derived ecosystem service deficit and planning priority model  
Prepared by: Abdullah Abdazeez Ayomide | Project year: 2025

Figure 8. Urban green-infrastructure planning-priority zones in Tartu.

| Priority class | Area (ha) | Area (km <sup>2</sup> ) | Share (%) |
|----------------|-----------|-------------------------|-----------|
| Very Low       | 1251.20   | 12.512                  | 32.18     |
| Low            | 321.01    | 3.210                   | 8.25      |
| Moderate       | 293.47    | 2.935                   | 7.55      |
| High           | 813.87    | 8.139                   | 20.93     |
| Very High      | 1209.17   | 12.092                  | 31.09     |

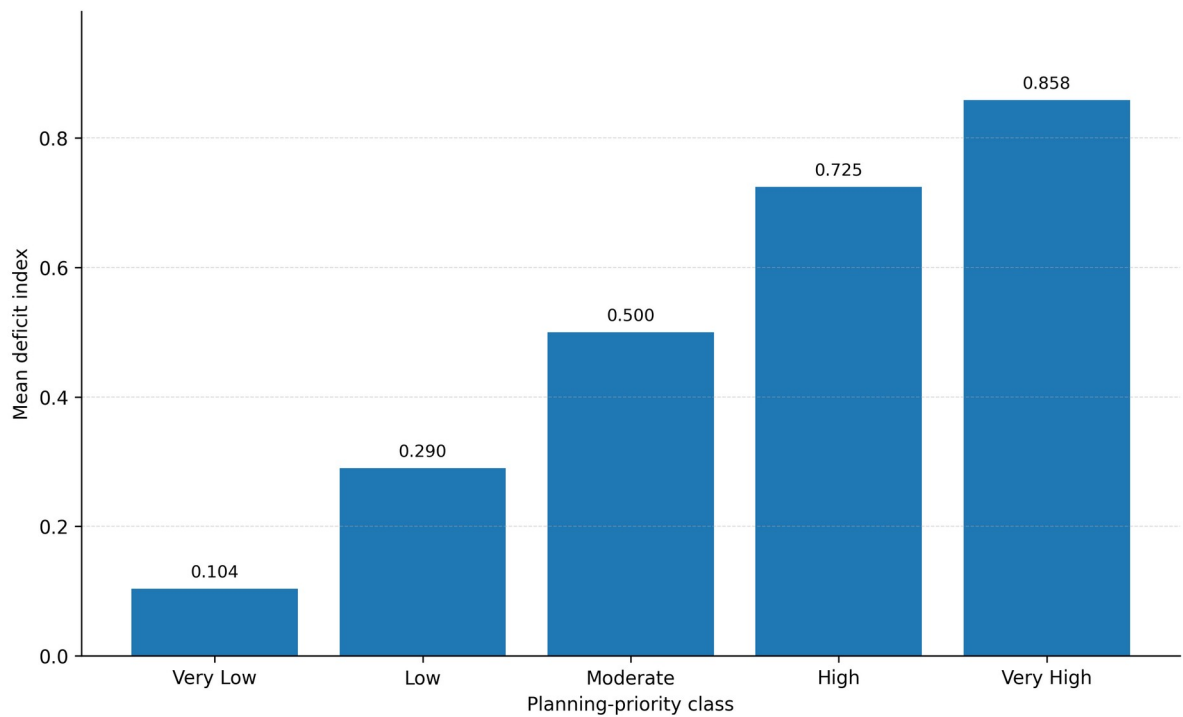
## Urban Green Infrastructure Planning-Priority Distribution



Mapped area: 38.887 km<sup>2</sup> | Higher classes indicate greater intervention urgency

Figure 9. Distribution of the five planning-priority classes.

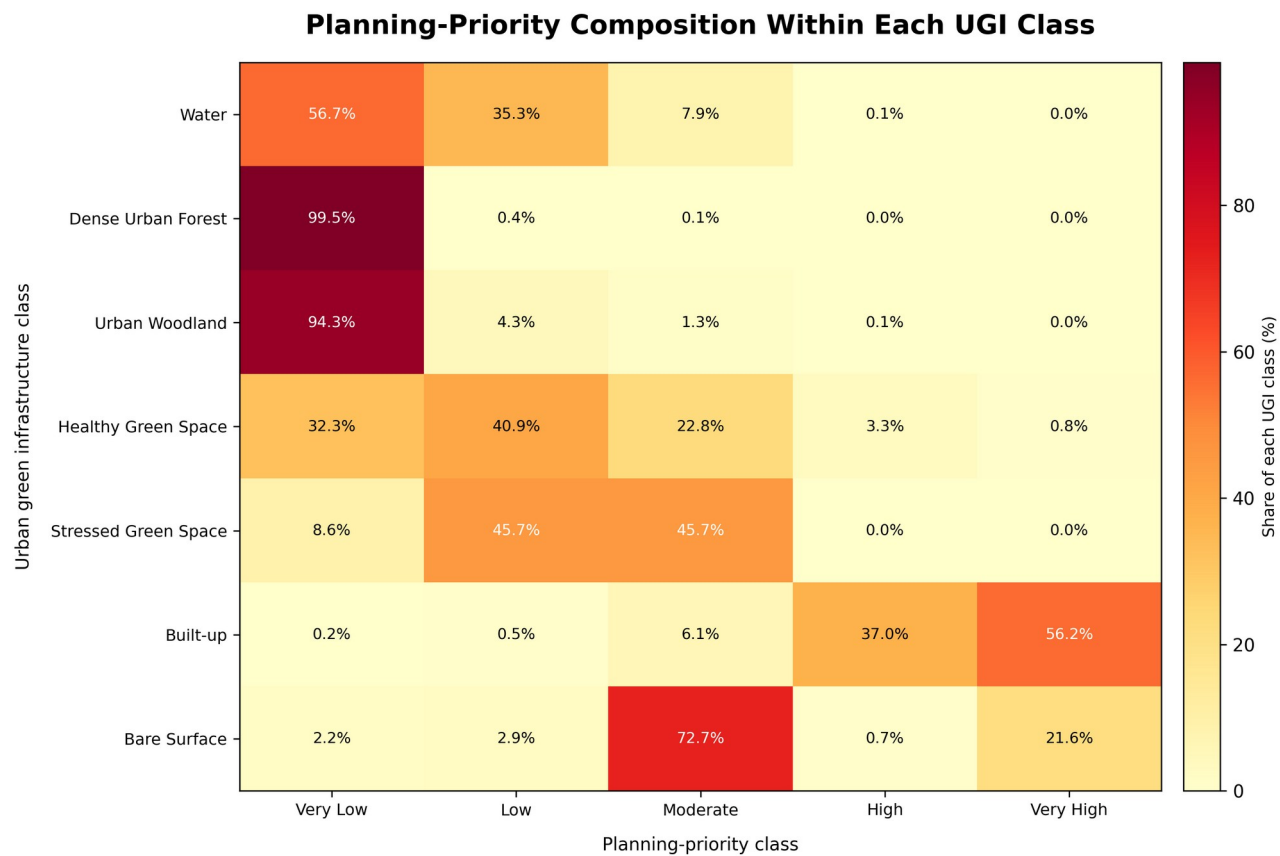
## Mean Ecosystem Service Deficit by Planning-Priority Class



Statistics calculated using cells with complete capacity, demand, deficit and priority data

Figure 10. Mean ecosystem-service deficit by planning-priority class.

5.6 UGI and Planning-Priority Cross-Classification



Calculated using the common comparison area of 38,444 km².

Figure 11. Planning-priority composition within each urban green-infrastructure class.

| UGI class            | High/Very High area (km²) | Share of class in High/Very High (%) | Share of urgent area (%) |
|----------------------|---------------------------|--------------------------------------|--------------------------|
| Water                | 0.001                     | 0.07                                 | 0.00                     |
| Dense Urban Forest   | 0.000                     | 0.00                                 | 0.00                     |
| Urban Woodland       | 0.002                     | 0.16                                 | 0.01                     |
| Healthy Green Space  | 0.269                     | 4.07                                 | 1.33                     |
| Stressed Green Space | 0.000                     | 0.00                                 | 0.00                     |
| Built-up             | 19.955                    | 93.14                                | 98.64                    |
| Bare Surface         | 0.003                     | 22.30                                | 0.02                     |

## 6. Results Interpretation

### 1. Urban Green Infrastructure Structure

The urban green infrastructure classification shows that Tartu's compact urban study area contains a substantial but spatially uneven green-infrastructure resource. Built-up land occupies 55.10% of the area, while vegetated UGI classes collectively account for 42.74%. Tree-based green infrastructure, comprising dense urban forest and urban woodland, covers 9.844 km<sup>2</sup> or 25.32% of the mapped area. Dense urban forest is the largest individual green-infrastructure class, occupying 8.549 km<sup>2</sup> or 21.99% of the study area. Healthy green space occupies 6.759 km<sup>2</sup> or 17.38%, while urban woodland accounts for 1.295 km<sup>2</sup> or 3.33%. Water covers 0.828 km<sup>2</sup> or 2.13%.

### 2. Ecosystem Service Capacity

The mean ecosystem-service capacity score is 36.71 on the project's 0–100 scale. The capacity distribution is strongly influenced by the contrast between vegetated and built-up surfaces. High-capacity areas are associated primarily with dense forest, woodland and healthy green space, while low-capacity values dominate extensively developed surfaces. The very high share of dense urban forest classified as Very Low planning priority (99.54%) confirms the strong service capacity assigned to this class. Similarly, 94.27% of urban woodland lies within the Very Low priority category.

### 3. Ecosystem Service Demand

The mean normalized ecosystem-service demand index is 0.379. Demand varies across the compact urban area, reflecting differences in urban intensity and the spatial concentration of environmental and population-related need. Demand data cover 38.444 km<sup>2</sup> of the study area, equivalent to approximately 98.86% of the authoritative boundary.

### 4. Ecosystem Service Deficit

The mean ecosystem-service deficit index is 0.514. The deficit surface represents the spatial mismatch between ecosystem-service capacity and demand. High values occur where demand is relatively strong but existing green-infrastructure capacity is limited. The internal model relationships are strong: capacity and deficit have a correlation of -0.9735, while demand and deficit have a correlation of 0.9588. These values confirm the mathematical consistency of the deficit model because deficit was derived from capacity and demand. They should not be interpreted as independent causal relationships.

### 5. Planning-Priority Distribution

High and Very High planning-priority zones jointly occupy 20.230 km<sup>2</sup> or 52.02% of the study area. Moderate-priority areas occupy 2.935 km<sup>2</sup> or 7.55%. Low and Very Low priority zones jointly cover 15.722 km<sup>2</sup> or 40.43%. The deficit gradient supports the priority classification. Mean deficit rises from 0.104 in Very Low priority areas to 0.290 in Low, 0.500 in Moderate, 0.725 in High and 0.858 in Very High priority areas.

### 6. Relationship Between UGI Classes and Planning Priority

The cross-classification reveals a pronounced division between developed and green surfaces. Built-up land dominates the highest-priority categories: 93.14% of built-up land is classified as High or Very High priority. Built-up areas also account for 98.64% of all High and Very High priority land. In contrast, dense urban forest and urban woodland are overwhelmingly concentrated in the Very Low priority category. Healthy green space has a more mixed profile: 73.17% lies within Low or Very Low priority zones, while only 4.07% occurs within High or Very High priority zones. Water surfaces also show a predominantly low-priority profile, with 92.08% classified as Low or Very Low priority. The results indicate that existing green infrastructure largely performs a protective ecosystem-service role, while the most significant service deficits are concentrated in extensively built-up parts of the urban area.

## 7. Planning Implications and Recommendations

### 1. Prioritize Built-Up Deficit Hotspots

The primary intervention focus should be the 19.955 km<sup>2</sup> of built-up land located within High and Very High priority zones. Because these surfaces comprise 98.64% of the total urgent area, strategies focused on developed land will address the largest share of the identified ecosystem-service deficit.

### 2. Protect Existing High-Capacity Green Infrastructure

Dense urban forest and urban woodland should be treated as critical green infrastructure assets. Their concentration in Very Low priority zones indicates that they provide substantial ecosystem-service capacity relative to local demand. Planning controls should therefore emphasize retention, connectivity and avoidance of unnecessary fragmentation.

### 3. Enhance Green Infrastructure Within Developed Areas

High-priority built-up zones are suitable locations for interventions that can be integrated into the existing urban fabric. Relevant measures include street-tree planting, vegetated corridors, pocket parks, green roofs, green walls, permeable surfaces and the greening of public institutional land.

### 4. Strengthen Green-Space Connectivity

Existing forests, woodland, healthy green spaces and water-related open spaces should be connected where spatial opportunities exist. Improved connectivity can support continuity between high-capacity areas and reduce isolated deficit hotspots.

### 5. Use Moderate-Priority Zones for Preventive Planning

The 2.935 km<sup>2</sup> of Moderate priority land should not be ignored. These areas represent transitional locations where targeted, lower-intensity intervention may prevent future escalation into High or Very High priority conditions.

### 6. Apply Context-Specific Intervention Design

The priority map identifies where intervention need is greatest, but it does not prescribe one uniform solution. Site-level planning should consider land ownership, current land use, development intensity, public-space availability, infrastructure constraints and implementation feasibility.

### 7. Integrate the Priority Surface Into Planning Decisions

The final planning-priority raster can support preliminary screening for urban development review, green-space investment, neighbourhood upgrading and ecosystem-service enhancement. It should be used as a strategic decision-support layer rather than as a substitute for detailed site assessment.

#### 7.1 Planning Action Register

| Order | Planning action                                   | Spatial evidence                                                                       | Recommended approach                                                                                              |
|-------|---------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1     | Target built-up High and Very High priority zones | 19.955 km <sup>2</sup> of built-up land falls within High or Very High priority zones. | Street trees, pocket parks, green roofs, vegetated corridors, permeable surfaces and institutional-land greening. |
| 2     | Protect dense urban forest                        | 99.54% of dense urban forest is classified as Very Low priority.                       | Retention, development buffers, fragmentation control and ecological-management measures.                         |
| 3     | Protect and connect urban woodland                | 94.27% of urban woodland lies in Very Low priority zones.                              | Improve connectivity between woodland, forest, parks and water-related green spaces.                              |
| 4     | Strengthen healthy green spaces                   | 4.07% of healthy green space occurs in High or Very High priority zones.               | Improve vegetation structure, accessibility, connectivity and ecological performance where local deficits remain. |
| 5     | Apply preventive action in Moderate zones         | Moderate priority areas occupy 2.935 km <sup>2</sup> .                                 | Use early-stage greening and development controls to prevent further ecosystem-service decline.                   |

## 8. Limitations

### 1. The ecosystem-service capacity, demand, deficit and priority outputs are

model-derived indicators. They represent comparative spatial conditions rather than direct measurements of every ecosystem service.

### 2. The demand raster contains limited NoData coverage. Cross-dataset statistics

therefore use a common valid area of 38.444 km<sup>2</sup>, while individual UGI and priority summaries use the full rasterized boundary area of 38.887 km<sup>2</sup>.

### 3. The priority classification depends on the indicators, normalization methods,

thresholds and weighting framework established during the analytical stages.

### 4. The results identify strategic planning priorities at the project raster

resolution. Detailed intervention design requires field verification and site-specific assessment.

### 5. The reported correlations are internal model relationships because the

ecosystem-service deficit was calculated from capacity and demand. They must not be presented as independent evidence of causation.

### 6. Small classes such as stressed green space and bare surface contain relatively

few pixels. Percentage interpretations for these classes should therefore be treated cautiously.

### 7. The outputs represent the project reference period and should be updated when

newer land-cover, population or planning data become available.

## 9. Conclusion

The analysis demonstrates that Tartu's compact urban area contains substantial green infrastructure, but ecosystem-service capacity and planning need are unevenly distributed. Vegetated UGI covers 42.74% of the mapped area, while built-up surfaces account for 55.10%.

More than half of the study area—52.02% or 20.230 km<sup>2</sup>—is classified as High or Very High planning priority. The urgent planning burden is concentrated almost entirely within built-up surfaces: 93.14% of built-up land lies in the two highest priority categories, and built-up areas constitute 98.64% of all High and Very High priority land.

The results support a two-part planning strategy. Existing high-capacity forests, woodlands and healthy green spaces should be protected and connected, while built-up deficit hotspots should receive targeted green-infrastructure interventions. The final priority map provides a consistent spatial framework for screening, investment prioritization and further site-level planning.

## 10. Technical Output Summary

| Output category             | Quantity | Formats      |
|-----------------------------|----------|--------------|
| Final maps                  | 5        | PNG and PDF  |
| Analytical charts           | 6        | PNG and PDF  |
| Final statistical tables    | 12       | CSV          |
| Report narrative components | 7        | TXT          |
| Authoritative mask          | 1        | GeoTIFF      |
| Final technical report      | 2        | DOCX and PDF |

All results presented in this report were derived from the validated final analytical outputs. The report-assembly stage did not alter or rerun the underlying spatial analysis.